

SGA 5 — Exposé I

Dualizing Complexes

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Introduction

This exposé is devoted to the biduality theorem in étale cohomology. The theory developed here and in SGA A XVIII (SGA A = SGA 4), for the étale topology and constructible torsion sheaves on preschemes, is formally very analogous to the one developed in HARTSHORNE [H] for the Zariski topology and coherent sheaves: the latter served, to a very large extent, as its model. But, whereas in the case of “continuous” coefficients the existence of dualizing complexes presents no difficulties, in the case of “discrete” coefficients, on the other hand, it is much more delicate to prove, and it seems to require resolution of singularities in an essential way. See however Deligne’s biduality theorem (SGA 4^{1/2}, Finiteness Theorem 4.3.).

Paragraph 1 contains the definition of dualizing complexes and the formal properties that follow from it, notably the interpretation of dualizing functors as “exchangers of functors.”

In paragraph 2, one shows that, on a connected locally noetherian prescheme, a dualizing complex is determined uniquely, up to a translation of degrees and tensoring by an invertible sheaf.

In paragraph 3, one comes to the central theorem of the exposé (3.4.1.), which asserts that, on a “good” regular prescheme X , for example on an excellent noetherian regular prescheme of characteristic zero¹, the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing.

Paragraph 4 gives the applications of the theory to the local duality theorem. The latter expresses that, on a “good” prescheme X endowed with a closed point x whose residue field is separably closed, and for a constructible sheaf F , one has a perfect pairing between $\underline{H}_x^i(F')$ and $\underline{H}_x^{-i}(F)$, where F' is the dual of F , i.e. $F' = R\underline{\mathrm{Hom}}(F, K_X)$, where K_X is a dualizing complex.

Finally, in paragraph 5, one proves directly that on a regular prescheme of dimension 1 the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing (n prime to the residual characteristics).

1 Definition and formal properties of dualizing complexes

We fix a coefficient ring $A = \mathbb{Z}/n\mathbb{Z}$. If X is a prescheme, we denote by A_X the constant sheaf on $X_{\text{ét}}$ with value A , and by $D(X)$ the derived category of the category of A_X -modules. We denote, on the other hand, by $D_c(X)$ the full triangulated subcategory of $D(X)$ formed by the complexes F such that $H^i(F)$ is a constructible A_X -module for every i . We put $D_c^*(X) = D^*(X) \cap D_c(X)$, where $*$ = $\emptyset$, $+$, $-$, or b .

¹This last restriction will become unnecessary once one has resolution of singularities à la Hironaka for these preschemes, and the “purity theorem.”

Definition 1.1

An A_X -module I is said to be *quasi-injective* if one has

$$\underline{\mathrm{Ext}}^i(F, I) = 0$$

for every *constructible* A_X -module F and every $i > 0$.

Remarks 1.2

- a) Contrary to what happens in the case of “continuous” coefficients (cf. [H] II.7.17), a quasi-injective sheaf is not injective in general. For example, let $X = \mathrm{Spec}(\mathbb{R})$ and $A = \mathbb{Z}/2\mathbb{Z}$: the sheaf A_X is quasi-injective, but is not injective, nor even of finite injective dimension, since $H^*(X; A_X) = H^*(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z}/2\mathbb{Z})$ is a polynomial algebra on one generator of dimension 1.
- b) Let I be a quasi-injective A_X -module. It is false in general that the relation $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > 0$, valid for constructible F , is valid for every F . Indeed, take $A = \mathbb{F}_\ell$. Let k be a field, $\bar{k}$ a separable closure of k , and $f : \bar{x} = \mathrm{Spec}(\bar{k}) \rightarrow X = \mathrm{Spec}(k)$ the canonical morphism, and suppose that, for every connected étale covering $X' \rightarrow X$, there exists a connected étale covering $X'' \rightarrow X'$ such that $[X'' : X']$ is divisible by ℓ (take, for example, $k = \mathbb{Q}$). Consider the exact sequence

$$(*) \quad 0 \longrightarrow A_X \xrightarrow{\varepsilon} f_* f^* A_X \longrightarrow F \longrightarrow 0$$

defined by the adjunction arrow ε . In view of the hypothesis, the section of $\mathrm{Ext}^1(F, A_X)$ defined by $(*)$ does not vanish above any U étale over X ; hence $\underline{\mathrm{Ext}}^1(F, A_X)_{\bar{x}} \neq 0$. But one checks trivially that A_X is quasi-injective.

Proposition 1.3 (cf. [H] 1.7.6)

Let X be a prescheme and $N \in \mathbb{Z}$. The following conditions on $K \in \mathrm{ob} D^+(X)$ are equivalent:

- (i) K is isomorphic, in $D(X)$, to a bounded complex I such that I^i is quasi-injective for every i and zero for $i > N$;
- (ii) for every $F \in \mathrm{ob} D_c^b(X)$ such that $H^i(F) = 0$ for $i < 0$, one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$;
- (iii) for every constructible A_X -module F , one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$.

Proof. (i) $\Rightarrow$ (ii). Let $F \in \mathrm{ob} D_c^b(X)$ be such that $H^i(F) = 0$ for $i < 0$, and let $I \in \mathrm{ob} D^+(X)$ be bounded and such that I^i is quasi-injective for every i and zero for $i > N$; we show that $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > N$. Arguing by induction on the number of indices i such that $H^i \neq 0$, one is reduced, by dévissage, to the case where $N = 0$ and I is concentrated in degree 0. One then has a biregular spectral sequence

$$E_2^{pq} = \underline{\mathrm{Ext}}^p(H^{-q}(F), I) \Rightarrow \underline{\mathrm{Ext}}^*(F, I).$$

Since I is quasi-injective, $E_2^{pq} = 0$ for $p > 0$ and every q , and hence one has $E_2^{0i} = \underline{\mathrm{Hom}}(H^{-i}(F), I) \xrightarrow{\sim} \underline{\mathrm{Ext}}^i(F, I)$, which gives the conclusion.

(ii) $\Rightarrow$ (iii) is trivial.

(iii) $\Rightarrow$ (i) We may suppose that K is bounded below and has injective components. The hypothesis, applied to $F = A_X$, implies that $H^i(K) = 0$ for $i > N$. We may therefore replace K by an isomorphic complex K' , bounded, and such that K'^i is injective for $i < N$ and zero for $i > N$. It is then observed that $K'^{!N}$ is quasi-injective, which completes the proof.

Remark 1.3.1

The writer does not know whether (1.3) is still true when, in condition (ii), the hypothesis “ $F \in \text{ob } D_c^b(X)$ ” is replaced by “ $F \in \text{ob } D_c^+(X)$ ”.

Definition 1.4

The *quasi-injective dimension* of K , denoted $\dim . q. inj(K)$, is the lower bound (in $\overline{\mathbb{R}}$) of the integers N satisfying the equivalent conditions of (1.3).

If K is of finite quasi-injective dimension, the functor $R \underline{\text{Hom}}(_, K) : D(X) \rightarrow D(X)$ induces a functor $D_c^b(X) \rightarrow D^b(X)$.

We shall see below that if X is smooth of dimension d over a field k and n is prime to the characteristic of k , then $\dim . q. inj(A_X) = 2d$.

The example of (1.2 a)) shows that a complex of finite quasi-injective dimension is not necessarily of finite injective dimension. Nevertheless one has:

Proposition 1.5

Let X be a locally noetherian prescheme and $K \in \text{ob } D^+(X)$. Suppose that there exists an integer N such that $\text{cd}_{\mathbf{n}}(X) \leq N$, where $\mathbf{n}$ is the set of prime divisors of n (notation of SGAA X 1). Then, for $N' \in \mathbb{Z}$, one has

$$\dim . q. inj(K) \leq N' \implies \dim . inj(K) \leq N + N'.$$

We shall use, in the proof, the following

Lemma 1.5.1

Let C be an abelian category satisfying (AB5) and possessing a family of generators $(U_i)_{i \in I}$. Let $K \in \text{ob } D^+(C)$ and $N \in \mathbb{Z}$. The following conditions are equivalent:

- (i) $\dim . inj(K) \leq N$;
- (ii) for every $i \in I$ and every quotient F of U_i , one has

$$\text{Ext}^n(F, K) = 0 \quad \text{for } n > N.$$

Proof. It suffices to prove (ii) $\Rightarrow$ (i). By a well-known argument (cf. for example [H] 1.7.6), one is reduced to showing that if $K' \in \text{ob}(C)$ is such that $\text{Ext}^1(F, K') = 0$ every time F is a quotient of a U_i , then K' is injective. In other words, one has to prove that, if every morphism from a subobject

V of a U_i into K' extends to U_i , then K' is injective. But this follows immediately from the proof of Lemma 1, p. 136 of (Tohoku).

Proof of (1.5). Suppose $\dim .q.inj(K) \leq N'$ and prove $\dim .inj(K) \leq N + N'$. Since the category of A_X -modules admits as generators the $A_{U,X}$ where U is étale of finite type over X , it suffices, by (1.5.1), to prove that $\text{Ext}^i(X; F, K) = 0$ for $i > N + N'$ when F is a quotient of such an $A_{U,X}$, hence constructible (SGAA IX 2.9). Consider the spectral sequence

$$E_2^{pq} = H^p(X; \underline{\text{Ext}}^q(F, K)) \Rightarrow \text{Ext}^*(X; F, K).$$

The hypothesis $\text{cd}_{\mathbf{n}}(X) \leq N$ (resp. $\dim .q.inj(K) \leq N'$) implies $E_2^{pq} = 0$ for $p > N$ (resp. $q > N'$). Hence $E_2^{pq} = 0$ for $p + q > N + N'$, whence the conclusion.

Remark 1.5.2

The hypothesis of (1.5) is verified (with $N = 2d$) if X is a prescheme of finite type and dimension d over a separably closed field k , with n prime to $\text{char}(k)$ (SGAA X 4).

Proposition 1.6

Let $f : X \rightarrow Y$ be a separated quasi-finite morphism of noetherian preschemes, and let $K \in \text{ob } D^+(Y)$. Then, for $N \in \mathbb{Z}$, one has

$$\dim .q.inj(K) \leq N \quad \implies \quad \dim .q.inj(R^!f(K)) \leq N.$$

Proof. According to the “Main theorem” (EGA IV 8.12.6), there exists a factorization of f as $f = uf'$, where f' is an open immersion and u is a finite morphism. We have $R^!f = R^!f' R^!u$, which reduces us to examining separately the case of an open immersion and that of a finite morphism. If f is an open immersion, the assertion is trivial (since every constructible sheaf on X is then induced by a constructible sheaf on Y). Suppose therefore that f is a finite morphism, and let F be a constructible A_X -module. We then have the duality isomorphisms (SGAA XVIII 3.1.9.6)

$$f_* \underline{\text{Ext}}^i(F, R^!f(K)) \xrightarrow{\sim} \underline{\text{Ext}}^i(f_* F, K).$$

Since $f_*(F)$ is constructible (SGAA IX 2.14), it follows that, if $\dim .q.inj(K) \leq N$, then $f_* \underline{\text{Ext}}^i(F, R^!f(K)) = 0$ for $i > N$, hence $\underline{\text{Ext}}^i(F, R^!f(K)) = 0$ for $i > N$ (SGAA VIII 5.5), as required.

Let now X be a prescheme, and let $K \in \text{ob } D^+(X)$. Denote by $\underline{D}_K$ the functor $R \underline{\text{Hom}}(-, K)$. We shall define, for $F \in \text{ob } D(X)$, a functorial homomorphism $F \rightarrow \underline{D}_K \underline{D}_K F$ (cf. [H] V.1.2). The construction which follows would be valid more generally on a ringed topos. We may first suppose that K is formed of injectives, which allows us to “remove the R ”. The identity morphism $\underline{\text{Hom}}(F, K) \rightarrow \underline{\text{Hom}}(F, K)$ defines, by the formula dear to Cartan, a homomorphism $F \otimes \underline{\text{Hom}}(F, K) \rightarrow K$, whence, by a second application of the said formula, a homomorphism $F \rightarrow \underline{\text{Hom}}(\underline{\text{Hom}}(F, K), K)$, which is the announced homomorphism, and which we baptize the canonical homomorphism. We say that the pair (F, K) is *bidualizing*, or *satisfies biduality*, or again that F is *reflexive* relative to K , if the canonical homomorphism $F \rightarrow \underline{D}_K \underline{D}_K F$ is an isomorphism.

From now on, unless expressly mentioned otherwise, all preschemes considered will be assumed lo

Definition 1.7

Let X be a prescheme. A complex $K \in \text{ob } D^+(X)$ is said to be *dualizing* if the following conditions are satisfied:

- (i) K is of finite quasi-injective dimension;
- (ii) for every $F \in \text{ob } D_c^-(X)$, one has $\underline{D}_K(F) \in \text{ob } D_c^+(X)$;
- (iii) every $F \in \text{ob } D_c^b(X)$ is reflexive relative to K .

Remark 1.8

- a) By a standard dévissage (truncating F and using the fact that $D_c(X)$ is triangulated), one sees that condition (ii) is equivalent to
(ii bis): for every constructible A_X -module F and every $i \in \mathbb{Z}$, $\underline{\text{Ext}}^i(F, K)$ is constructible.
- b) Condition (ii bis) implies $K \in \text{ob } D_c^+(X)$. Contrary to the example of coherent sheaves (EGA 0_{III} 12.3.3), the converse is perhaps not automatically verified. It is however verified (see no. 3) in the “good” cases, for example when one has on X resolution of singularities and purity (in particular if X is an excellent prescheme of characteristic zero).
- c) By dévissage on F , one sees that condition (iii) is equivalent to (iii bis): every constructible A_X -module F is reflexive relative to K .
- d) If $K \in \text{ob } D^+(X)$ satisfies (i) and (ii) (resp. is dualizing), the functor $\underline{D}_K$ induces a functor (resp. an equivalence of categories) $(D_c^b(X))^\circ \rightarrow D_c^b(X)$.
- e) If $K \in \text{ob } D^+(X)$ is of finite *injective* dimension (cf. 1.5.2) and satisfies (ii), then, for every $F \in \text{ob } D_c(X)$, one has $\underline{D}_K(F) \in D_c(X)$, and $\underline{D}_K$ induces a functor $(D_c^+(X))^\circ \rightarrow D_c^-(X)$.
If moreover K is dualizing, every $F \in \text{ob } D_c(X)$ is reflexive relative to K , and $\underline{D}_K$ induces equivalences of categories

$$(D_c(X))^\circ \xrightarrow{\sim} D_c(X), \quad (D_c^-(X))^\circ \xrightarrow{\sim} D_c^+(X), \quad (D_c^+(X))^\circ \xrightarrow{\sim} D_c^-(X).$$

It is not known whether this conclusion remains valid without the restriction that K be of finite injective dimension.